1.py - C:/Users/kavip/AppData/Local/Programs/Python/Python314/1.py (3.14.1)

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```
n = int(input("Enter number of vertices: "))
e = int(input("Enter number of edges: "))

graph = {}

for i in range(1, n + 1):
    graph[i] = []

print("Enter edges:")

for i in range(e):
    u, v = map(int, input().split())
    graph[u].append(v)
    graph[v].append(u)

start = int(input("Enter starting vertex: "))

visited = set()

def dfs(node):
    visited.add(node)
    print(node, end=" ")

    for neighbour in graph[node]:
        if neighbour not in visited:
            dfs(neighbour)

print("DFS Traversal:", end=" ")
dfs(start)
```

Ln: 30 Col: 0

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IDLE Shell 3.14.1

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```
Python 3.14.1 (tags/v3.14.1:57e0d17, Dec  2 2025, 14:05:07) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/kavip/AppData/Local/Programs/Python/Python314/1.py =====
Enter number of vertices: 8
Enter number of edges: 8
Enter edges:
1 2
1 3
2 5
2 6
3 7
3 4
4 8
7 8
Enter starting vertex: 1
DFS Traversal: 1 2 5 6 3 7 8 4
>>>
```

Ln: 18 Col: 0

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IDLE Shell 3.14.1

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```
Python 3.14.1 (tags/v3.14.1:57e0d17, Dec  2 2025, 14:05:07) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/kavip/AppData/Local/Programs/Python/Python314/1.py =====
Enter number of vertices: 10
Enter number of edges: 9
Enter edges:
1 2
1 7
2 3
2 6
7 8
7 10
3 4
3 5
8 9
Enter starting vertex: 1
DFS Traversal: 1 2 3 4 5 6 7 8 9 10
>>>
```

Ln: 19 Col: 0

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IDLE Shell 3.14.1

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```
Python 3.14.1 (tags/v3.14.1:57e0d17, Dec  2 2025, 14:05:07) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/kavip/AppData/Local/Programs/Python/Python314/1.py =====
Enter number of vertices: 7
Enter number of edges: 6
Enter edges:
A B
A C
B D
B E
C F
C G
Enter starting vertex: A
DFS Traversal: A B D E C F G
>>>
```

Ln: 16 Col: 0

33° Search ENG IN 13:15 12-10-2026